

openapi: 3.1.0

info:

title: Cloud Health Office – Payer-to-Payer Data Exchange API

version: 4.0.0

description: |

FHIR R4 Payer-to-Payer Data Exchange API for CMS-0057-F compliance.

When a member switches health plans, CMS-0057-F requires the new plan to request clinical and claims data from the previous plan – and the previous plan must provide it. This API implements both sides of that exchange.

****Workflow:****

1. New payer calls `\$member-match` to identify the member in the old payer's
2. New payer verifies member consent via the Consent endpoint
3. New payer calls `\$everything` or bulk export to retrieve all member data
4. Old payer packages FHIR resources (Patient, Claims, EOBs, Conditions, etc. as NDJSON and uploads to Azure Data Lake / Blob Storage
5. New payer imports and reconciles (deduplication, conflict resolution)

****Key Features:****

- FHIR \$member-match operation for cross-payer member identification
- Bulk FHIR export/import with NDJSON format
- Member consent validation (active, scoped, time-limited)
- Azure Service Bus for async workflow orchestration
- Azure Data Lake for bulk data staging
- Deduplication and conflict resolution on import
- US Core profile validation

****References:****

- CMS-0057-F Payer-to-Payer Data Exchange requirements
- Da Vinci PDex Implementation Guide
- FHIR Bulk Data Access IG (Flat FHIR)

contact:

name: Aureliaware, Inc.

url: <https://cloudhealthoffice.com>

license:

name: Apache 2.0

url: <https://www.apache.org/licenses/LICENSE-2.0>

servers:

- url: <https://api.cloudhealthoffice.com/fhir/r4>
description: Production
- url: <https://sandbox.cloudhealthoffice.com/fhir/r4>
description: Sandbox

- url: http://localhost:3000/fhir/r4
description: Local development

security:

- oauth2_system: [system/*.read, system/Patient.read]

tags:

- name: Member Match
description: Cross-payer member identification
- name: Consent
description: Member consent for payer-to-payer data sharing
- name: Export
description: Bulk data export (responding to data requests)
- name: Import
description: Bulk data import (receiving transferred data)

paths:

— Member Match —————

/Patient/\$member-match:

post:

operationId: memberMatch

summary: Match a member across payer systems

description: |

The requesting (new) payer submits member demographics, coverage info, and consent to the responding (old) payer. The old payer returns the matched Patient resource with internal identifiers.

This is the first step in any payer-to-payer exchange — you must identify the member before requesting their data.

tags: [Member Match]

requestBody:

required: true

content:

application/fhir+json:

schema:

\$ref: '#/components/schemas/MemberMatchParameters'

example:

resourceType: Parameters

parameter:

- name: MemberPatient

resource:

resourceType: Patient

identifier:

- system: "http://hl7.org/fhir/sid/us-medicare"
value: "1EG4-TE5-MK72"

```

      name:
        - family: "Smith"
          given: ["Jane", "Marie"]
        gender: female
        birthDate: "1985-03-15"
    - name: CoverageToMatch
      resource:
        resourceType: Coverage
        identifier:
          - system: "https://old-payer.com/coverage-id"
            value: "COV-OLD-12345"
        subscriberId: "MBR-OLD-001"
        payor:
          - reference: "Organization/old-payer"
        period:
          start: "2023-01-01"
          end: "2025-12-31"
    - name: CoverageToLink
      resource:
        resourceType: Coverage
        subscriberId: "MBR-NEW-001"
        payor:
          - reference: "Organization/new-payer"
        period:
          start: "2026-01-01"
    - name: Consent
      resource:
        resourceType: Consent
        status: active
        scope:
          coding:
            - system: "http://terminology.hl7.org/CodeSystem/consents"
              code: "patient-privacy"
        patient:
          reference: "Patient/jane-smith"
responses:
  '200':
    description: Member matched successfully
    content:
      application/fhir+json:
        schema:
          type: object
        example:
          resourceType: Parameters
          parameter:
            - name: MemberIdentifier
              valueIdentifier:

```

```

      system: "https://cloudhealthoffice.com/member-id"
      value: "pat-12345"
    - name: MemberPatient
      resource:
        resourceType: Patient
        id: "pat-12345"
        identifier:
          - system: "https://cloudhealthoffice.com/member-id"
            value: "pat-12345"
        name:
          - family: "Smith"
            given: ["Jane", "Marie"]
'422':
  description: No match found or ambiguous match
  content:
    application/fhir+json:
      schema:
        $ref: '#/components/schemas/OperationOutcome'
      example:
        resourceType: OperationOutcome
        issue:
          - severity: error
            code: not-found
            diagnostics: "No unique member match found for the provided d

```

— Consent Registration —————

```

/Consent:
  post:
    operationId: registerConsent
    summary: Register member consent for payer-to-payer exchange
    description: |
      The requesting payer registers member consent before initiating
      a data export. Consent must be active, scoped to specific resource
      types, and may have an expiration date.

      Consent is validated on every export request.
    tags: [Consent]
    requestBody:
      required: true
    content:
      application/json:
        schema:
          $ref: '#/components/schemas/MemberConsent'
        example:
          patientId: "pat-12345"
          sourcePayerId: "payer-current"

```

```
    targetPayerId: "payer-new"
    status: "active"
    authorizedResourceTypes:
      - Patient
      - Claim
      - ExplanationOfBenefit
      - Encounter
      - Condition
      - Observation
      - Coverage
    consentDate: "2026-02-16T00:00:00Z"
    expirationDate: "2026-08-16T00:00:00Z"
```

```
responses:
  '201':
    description: Consent registered
  '400':
    description: Invalid consent record
```

/Consent/{patientId}/{targetPayerId}:

```
get:
  operationId: checkConsent
  summary: Verify consent for a specific payer exchange
  tags: [Consent]
  parameters:
    - name: patientId
      in: path
      required: true
      schema:
        type: string
    - name: targetPayerId
      in: path
      required: true
      schema:
        type: string
  responses:
    '200':
      description: Consent status
      content:
        application/json:
          schema:
            type: object
            properties:
              hasConsent:
                type: boolean
              authorizedResourceTypes:
                type: array
                items:
```

```
        type: string
expirationDate:
    type: string
    format: date-time
```

— Bulk Export (Responding Payer) —————

/Group/{groupId}/\$export:

post:

operationId: initiateBulkExport

summary: Initiate bulk data export for member transfer

description: |

The responding (old) payer packages member data as NDJSON files and stages them for download. Consent is validated for every member in the request before export begins.

This is an async operation – the response includes a polling URL to check export status.

tags: [Export]

parameters:

- name: groupId

in: path

required: true

schema:

type: string

description: Group identifier (or use `all` for single-member export)

requestBody:

required: true

content:

application/json:

schema:

\$ref: '#/components/schemas/BulkExportRequest'

example:

sourcePayerId: "payer-current"

targetPayerId: "payer-new"

memberIds:

- "pat-12345"

- "pat-67890"

resourceTypes:

- Patient

- Claim

- ExplanationOfBenefit

- Encounter

- Condition

since: "2020-01-01T00:00:00Z"

responses:

'202':

```
description: Export accepted – poll for status
headers:
  Content-Location:
    description: URL to poll for export status
    schema:
      type: string
      format: uri
content:
  application/json:
    schema:
      $ref: '#/components/schemas/BulkExportResult'
    example:
      exportId: "exp-20260216-001"
      status: "queued"
      requestedAt: "2026-02-16T10:00:00Z"
      patientsRequested: 2
      patientsWithConsent: 2
      patientsWithoutConsent: 0
      message: "Export queued. Poll Content-Location for status."
'400':
  description: Invalid request
'403':
  description: No members have valid consent
```

/export-status/{exportId}:

```
get:
  operationId: getExportStatus
  summary: Check bulk export status
  tags: [Export]
  parameters:
    - name: exportId
      in: path
      required: true
      schema:
        type: string
  responses:
    '200':
      description: Export complete – download links available
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/BulkExportResult'
          example:
            exportId: "exp-20260216-001"
            status: "completed"
            completedAt: "2026-02-16T10:02:30Z"
            resourcesExported: 47
```

```

        outputFiles:
          - type: Patient
            url: "https://storage.cloudhealthoffice.com/exports/exp-001/P
            count: 2
          - type: Claim
            url: "https://storage.cloudhealthoffice.com/exports/exp-001/C
            count: 18
          - type: ExplanationOfBenefit
            url: "https://storage.cloudhealthoffice.com/exports/exp-001/E
            count: 15
          - type: Condition
            url: "https://storage.cloudhealthoffice.com/exports/exp-001/C
            count: 12
      '202':
        description: Export still in progress

# — Bulk Import (Requesting Payer) —————

/import:
  post:
    operationId: initiateBulkImport
    summary: Import member data from another payer
    description: |
      The requesting (new) payer imports NDJSON files received from
      the responding payer. Handles deduplication, conflict resolution,
      and US Core profile validation.

      Duplicate detection uses resource type + identifier system + value.
    tags: [Import]
    requestBody:
      required: true
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/BulkImportRequest'
          example:
            sourcePayerId: "payer-old"
            blobUrls:
              - "https://storage.cloudhealthoffice.com/exports/exp-001/Patient.
              - "https://storage.cloudhealthoffice.com/exports/exp-001/Claim.nd
              - "https://storage.cloudhealthoffice.com/exports/exp-001/Explanat
              - "https://storage.cloudhealthoffice.com/exports/exp-001/Conditio
    responses:
      '202':
        description: Import accepted
        content:
          application/json:

```



```

      schema:
        $ref: '#/components/schemas/BulkImportResult'
    example:
      importId: "imp-20260216-001"
      status: "processing"
      totalResources: 47
      imported: 0
      duplicatesSkipped: 0
      errors: 0
  '400':
    description: Invalid request

```

/import-status/{importId}:

```

get:
  operationId: getImportStatus
  summary: Check bulk import status
  tags: [Import]
  parameters:
    - name: importId
      in: path
      required: true
      schema:
        type: string
  responses:
    '200':
      description: Import status
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/BulkImportResult'
          example:
            importId: "imp-20260216-001"
            status: "completed"
            completedAt: "2026-02-16T10:05:00Z"
            totalResources: 47
            imported: 42
            duplicatesSkipped: 5
            errors: 0

```

components:

```

schemas:
  MemberMatchParameters:
    type: object
  properties:
    resourceType:
      type: string
    const: Parameters

```

```
parameter:
  type: array
  items:
    type: object
    properties:
      name:
        type: string
        enum: [MemberPatient, CoverageToMatch, CoverageToLink, Consent]
      resource:
        type: object
```

```
MemberConsent:
  type: object
  required: [patientId, sourcePayerId, targetPayerId, status, authorizedResou
properties:
  patientId:
    type: string
  sourcePayerId:
    type: string
  targetPayerId:
    type: string
  status:
    type: string
    enum: [active, inactive, revoked]
  authorizedResourceTypes:
    type: array
    items:
      type: string
  consentDate:
    type: string
    format: date-time
  expirationDate:
    type: string
    format: date-time
```

```
BulkExportRequest:
  type: object
  required: [sourcePayerId, targetPayerId, memberIds]
properties:
  sourcePayerId:
    type: string
  targetPayerId:
    type: string
  memberIds:
    type: array
    items:
      type: string
```

```
resourceTypes:
  type: array
  items:
    type: string
  default: [Patient, Claim, ExplanationOfBenefit, Encounter, Condition, C
since:
  type: string
  format: date-time
  description: Only export resources updated after this date
```

```
BulkExportResult:
  type: object
  properties:
    exportId:
      type: string
    status:
      type: string
      enum: [queued, processing, completed, failed]
    requestedAt:
      type: string
      format: date-time
    completedAt:
      type: string
      format: date-time
    patientsRequested:
      type: integer
    patientsWithConsent:
      type: integer
    patientsWithoutConsent:
      type: integer
    resourcesExported:
      type: integer
    outputFiles:
      type: array
      items:
        type: object
        properties:
          type:
            type: string
          url:
            type: string
            format: uri
          count:
            type: integer
    message:
      type: string
```

BulkImportRequest:

```
  type: object
  required: [sourcePayerId, blobUrls]
  properties:
    sourcePayerId:
      type: string
    blobUrls:
      type: array
      items:
        type: string
        format: uri
```

BulkImportResult:

```
  type: object
  properties:
    importId:
      type: string
    status:
      type: string
      enum: [processing, completed, failed]
    completedAt:
      type: string
      format: date-time
    totalResources:
      type: integer
    imported:
      type: integer
    duplicatesSkipped:
      type: integer
    errors:
      type: integer
```

OperationOutcome:

```
  type: object
  properties:
    resourceType:
      type: string
      const: OperationOutcome
    issue:
      type: array
      items:
        type: object
        properties:
          severity:
            type: string
          code:
            type: string
```

```
      diagnostics:
        type: string

securitySchemes:
  oauth2_system:
    type: oauth2
    description: |
      System-level OAuth 2.0 (Backend Services / Client Credentials).
      Payer-to-payer exchange uses system-level tokens, not user tokens.
      The requesting payer authenticates with its client credentials.
    flows:
      clientCredentials:
        tokenUrl: https://login.cloudhealthoffice.com/oauth2/token
        scopes:
          system/*.read: Read all resources (system-level)
          system/Patient.read: Read patient demographics
          system/Claim.read: Read claims
          system/ExplanationOfBenefit.read: Read EOBs
```